



Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BNR:** Banca Națională a României — *the National Bank of Romania*
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CF:** Cornish–Fisher (expansion)
- ▣ **CI:** Confidence Interval
- ▣ **COVID-19:** COronaVirus Disease 2019
- ▣ **DM:** Diebold–Mariano (test)
- ▣ **ES:** Expected Shortfall
- ▣ **ETF:** Exchange-Traded Fund
- ▣ **EVT:** Extreme Value Theory
- ▣ **FHS:** Filtered Historical Simulation
- ▣ **FRTB:** Fundamental Review of the Trading Book
- ▣ **FZ0:** Fissler–Ziegel loss, zero-homogeneous version
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPD:** Generalised Pareto Distribution
- ▣ **HAC:** Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HS:** Historical Simulation
- ▣ **LGD:** Loss Given Default
- ▣ **MAR33:** Basel Framework, chapter MAR33 (internal models approach: capital requirements calculation)
- ▣ **OMV:** Österreichische Mineralölverwaltung — *Austrian Mineral Oil Administration (OMV group)*
- ▣ **QML:** Quasi-Maximum Likelihood
- ▣ **RMSFE:** Root Mean Squared Forecast Error
- ▣ **SAV:** Symmetric Absolute Value (CAViaR specification)
- ▣ **SE:** Standard Error
- ▣ **US:** United States

Today's Question and Seminar Structure

- Question of the seminar
 - ▶ **how large are tomorrow's tail losses, and how precisely can we tell?**
- Outputs to produce
 - ▶ the answers to the A exercises
 - ▶ one table and one figure per B exercise
 - ▶ the corrected C1 code
 - ▶ one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#)
 - ▶ each task names its notebook section
- Exercises, types and order: the Exercise Map (next slide)
 - ▶ B4, A5 and B5 use extreme value theory and Euler risk contributions, developed in Lecture 7: solve them after the lecture
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Worked example
A1, A2	VaR and ES under the Normal and the Student- <i>t</i> distributions	Solved	–
A3	VaR and ES of a discrete default loss	Solved	–
A4	when VaR punishes diversification	Proposed	A3
B1	historical VaR and ES on BET, with bootstrap intervals	Solved	–
B2	how precise is an ES estimate?	Proposed	B1
B3	tomorrow's filtered VaR for Bitcoin	Proposed	B1
C1	what is wrong in an AI answer?	Proposed	B1
B4	a GPD tail for BET (after the lecture)	Solved	–
A5, B5	risk contributions of two assets and of a four-asset portfolio (after the lecture)	Solved; Proposed	A5
Extension packs	A2, A4, A5, B1, B3, B4, B5 Extended; A6–A8, B6, B7; projects C2, C3 (optional, after the seminar)	Solved and Proposed	named on each task

- ▣ **[Solved]**: full solution in the slides and the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class; **[Open]**: a project question with no single correct answer (C2, C3)
- ▣ Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: AI critique and open projects

Data Used

Series	Price used	Sample	Calendar	Used in
BET	daily close	2000–2026	weekdays	B1, B2, B4, C1
S&P 500	daily close	2000–2026	US trading days	B2
Bitcoin	daily close (USD)	2014–2026	every day	B3
SPY, TLT, GLD	adjusted close	2014–2026	common days with Bitcoin	B5
Gold; EUR/RON	gold: daily close (USD per ounce); EUR/RON: BNR reference rate	2000–2026; 2005–2026	weekdays; Romanian working days	B6
BVB blue chips	Banca Transilvania, OMV Petrom and six other large BVB stocks, adjusted close	2016–2026	BVB trading days	C2

- Daily prices of the course data, one series per asset (field used: close or adjusted close, as in the table); all samples end on 18 September 2026
- EUR/RON: the official BNR reference rate, not the EUR/RON series of the market-data provider (it contains erroneous quotes, see Seminar 0)
- Two or more assets: align the *prices* on common trading days first, then compute returns

Setup: Run and Verify

```
import numpy as np, pandas as pd
URL = ('https://raw.githubusercontent.com/'
      'danpele/MFM/main/data/market/')
px = pd.read_csv(URL + 'BET.csv', index_col='date',
                 parse_dates=True)['close']
px = px.loc['2000-01-01':'2026-09-18']
px = px[px.index.dayofweek < 5]
r = 100 * np.log(px).diff().dropna()    # % log returns
L = -r                                  # losses
print(px.head(3)); print(len(r), r.index[-1])
```

- Code from the notebook section “Data”
- Expected output: $n = 6\,670$ returns, last date 2026-09-18

date	close	r_t (%)
2000-01-05	460.67	–
2000-01-06	484.04	4.95
2000-01-07	526.34	8.38

- Last close: 32 757.81 on 2026-09-18
- If your count or first value differs, check the series, the price column (close or adjusted close) and the weekday filter
- Other series: change only the series and the price column (slide “Data Used”)

What You Need for Today (1/21): VaR and ES

▣ Details: [Chapter 7](#)

▣ Return X , loss $L = -X$; the level α is a **tail probability**: VaR 1%, ES 2.5%

▣ **Value-at-Risk (VaR)**: the loss exceeded with probability α , i.e. minus the α -quantile of the return

$$\text{VaR}_\alpha = -q_\alpha(X) = q_{1-\alpha}(L), \quad q_\alpha(X) = \inf\{x : P(X \leq x) > \alpha\}$$

► $q_\alpha(X)$: the α -quantile, the smallest x (infimum) at which the distribution function $P(X \leq x)$ exceeds α ; VaR says how often a loss is exceeded, not by how much

▣ **Expected Shortfall (ES)**: the average loss in the worst α of outcomes

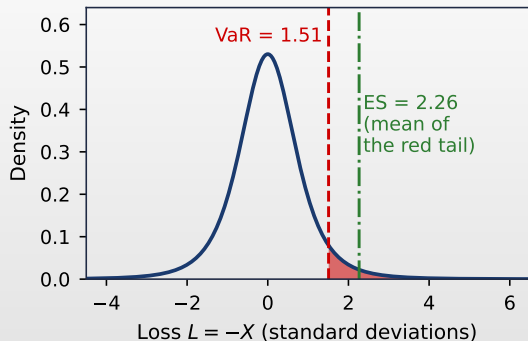
$$\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha] \quad (\text{continuous } L)$$

► An average of the VaRs at all levels $u \leq \alpha$, so $\text{ES}_\alpha \geq \text{VaR}_\alpha$; $\mathbb{E}[\cdot \mid \cdot]$: the mean over the outcomes that satisfy the condition

► Example: ten equally likely losses $-2, -1, 0, 0, 1, 1, 2, 3, 5, 9$, $\alpha = 20\%$: $P(L \geq 5) = 20\%$ is not above α , $P(L \geq 3) = 30\%$ is, so $\text{VaR}_{20\%} = 3$; the worst 20% are $\{9, 5\}$, so $\text{ES}_{20\%} = 7$

► Regulation: the Basel market-risk rules replaced VaR 1% by ES 2.5% [Basel Committee \(2019\)](#)

What You Need for Today (2/21): VaR and ES on a Loss Density



- Loss density, Student- $t(4)$, sd 1
- Tail, probability $\alpha = 5\%$

- A loss with a Student- $t(4)$ distribution scaled to standard deviation 1; tail probability $\alpha = 5\%$ in red
- VaR: where the red tail starts, $P(L > \text{VaR}_\alpha) = \alpha$
 - ▶ Here $\text{VaR}_{5\%} = 1.51$ standard deviations
- ES: the centre of mass of the red tail, so always to the right of VaR
 - ▶ Here $\text{ES}_{5\%} = 2.26$; two tails with the same VaR can have very different ES
- Signs: VaR and ES are reported as **positive** numbers when they are losses

What You Need for Today (3/21): Normal VaR and ES

- Normal returns: $X = \mu + \sigma Z$, $Z \sim N(0, 1)$

$$\text{VaR}_\alpha = -(\mu + \sigma z_\alpha), \quad \text{ES}_\alpha = -\mu + \sigma \frac{\varphi(z_\alpha)}{\alpha}$$

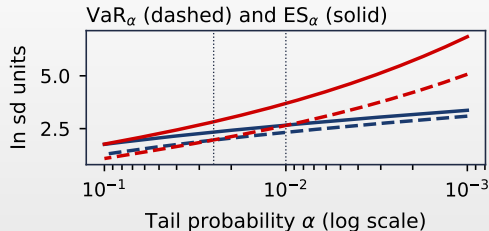
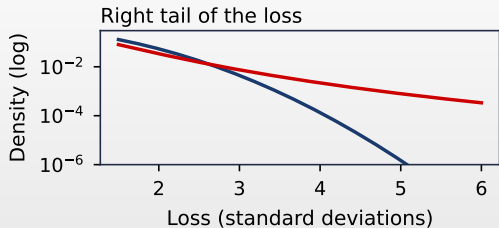
- ▶ μ : mean; σ : standard deviation; Φ , φ : distribution function and density of $N(0, 1)$; $z_\alpha = \Phi^{-1}(\alpha) < 0$: its α -quantile
- ▶ VaR lies $|z_\alpha|$ standard deviations below the mean; ES uses the larger factor $\varphi(z_\alpha)/\alpha$, the mean of $-Z$ over the tail (A1 derives it)
- ▶ $z_{1\%} = -2.326$, $\varphi(z_{1\%})/0.01 = 2.665$; $z_{2.5\%} = -1.960$, $\varphi(z_{2.5\%})/0.025 = 2.338$; with $\mu = 0$, $\sigma = 1\%$: VaR 1% = 2.33%, ES 2.5% = 2.34%

- Student- t returns: $X = m + sT$, $T \sim t_\nu$ with density g_ν and quantile $t_\alpha = t_\nu^{-1}(\alpha)$

$$\text{VaR}_\alpha = -(m + s t_\alpha), \quad \text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1} \quad (\nu > 1)$$

- ▶ m : location; $s > 0$: scale; ν : degrees of freedom, smaller ν = heavier tails; $\text{Var}(T) = \nu/(\nu - 2)$ for $\nu > 2$, so a standard deviation σ needs $s = \sigma \sqrt{(\nu - 2)/\nu}$ (A2)

What You Need for Today (4/21): Normal Against Student-t



— Normal, sd 1 — Student- $t(4)$, sd 1

- Both distributions have standard deviation 1; left: the right tail of the loss on a log scale, where the Student- $t(4)$ density is many times higher beyond 3 standard deviations
- Right: VaR (dashed) and ES (solid) in standard deviations, for α from 10% to 0.1%; dotted: $\alpha = 2.5\%$ and 1%
 - At moderate levels the two models are close (for α near 5% the Student- t VaR is even smaller); the gap opens deep in the tail, and faster for ES than for VaR

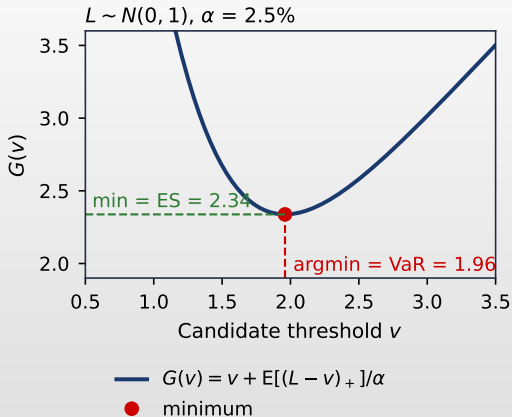
What You Need for Today (5/21): Coherent Risk Measures

- **Coherent risk measure** ρ Artzner et al. (1999): a rule that maps a loss L to the capital it needs and satisfies four axioms

$$\begin{aligned} L_1 \leq L_2 &\Rightarrow \rho(L_1) \leq \rho(L_2), & \rho(L + c) &= \rho(L) + c, \\ \rho(\lambda L) &= \lambda \rho(L) \ (\lambda > 0), & \rho(L_1 + L_2) &\leq \rho(L_1) + \rho(L_2) \end{aligned}$$

- ▶ Monotone (larger losses, more capital); translation invariant (a sure loss c adds c); positively homogeneous (double the position, double the capital)
 - ▶ **Subadditive**: merging two positions L_1, L_2 never needs more capital than holding them apart, so diversification is never penalised
- ES is coherent; VaR satisfies the first three axioms, but can fail subadditivity
 - ▶ Typical cases: discrete losses (defaults, A4) and very heavy tails; for jointly Normal losses VaR is subadditive

What You Need for Today (6/21): ES as a Minimum

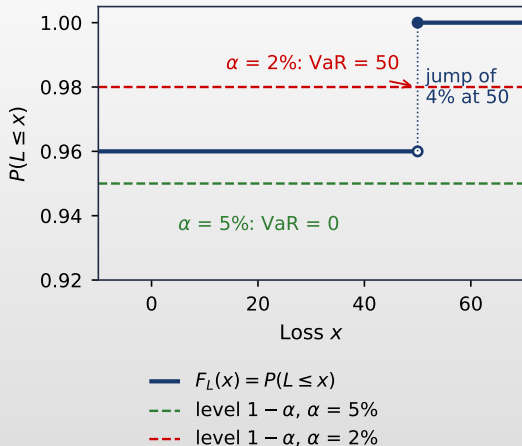


- **Rockafellar–Uryasev** [Rockafellar & Uryasev \(2000\)](#): ES is the minimum of a convex function of a candidate threshold v

$$G_\alpha(v) = v + \frac{\mathbb{E}[(L - v)_+]}{\alpha}, \quad \text{ES}_\alpha = \min_v G_\alpha(v)$$

- ▶ $(x)_+ = \max(x, 0)$; $\mathbb{E}[(L - v)_+]$: mean loss in excess of v (zero below v)
- ▶ Chart ($L \sim N(0, 1), \alpha = 2.5\%$): the minimum is reached at $v = \text{VaR}_\alpha = 1.96$ and equals $\text{ES}_\alpha = 2.34$
- Uses: ES estimation as an optimisation (A7), ES contributions by differentiation (A5 extended)

What You Need for Today (7/21): Losses with an Atom

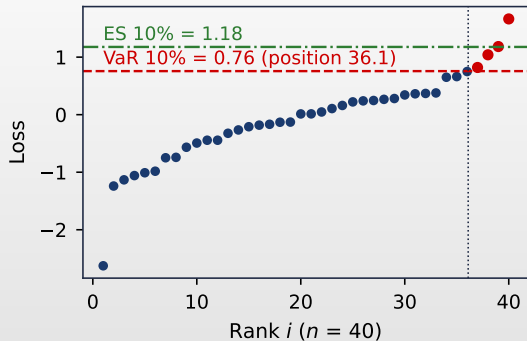


- An atom: a loss value with positive probability, e.g. a default; the distribution function jumps there
- VaR from the infimum definition: the first x where $P(L \leq x) \geq 1 - \alpha$
 - ▶ Chart (loss 50 with probability 4%): $\alpha = 5\%$ gives $\text{VaR} = 0$; $\alpha = 2\%$ gives $\text{VaR} = 50$
- ES must fill a tail of probability exactly α [Acerbi & Tasche \(2002\)](#)

$$\text{ES}_\alpha = \frac{1}{\alpha} \left(\mathbb{E}[L \mathbf{1}\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha (\alpha - P(L > \text{VaR}_\alpha)) \right)$$

- ▶ $\mathbf{1}\{\cdot\}$: 1 if the condition holds, 0 otherwise; first term: losses strictly beyond VaR; second: the missing tail probability, placed at VaR
- ▶ Chart, $\alpha = 5\%$: $(50 \times 0.04 + 0 \times 0.01) / 0.05 = 40$

What You Need for Today (8/21): Historical Simulation



- sorted losses $L_{(1)} \leq \dots \leq L_{(n)}$
- the $k = 4$ losses $\geq \text{VaR}$

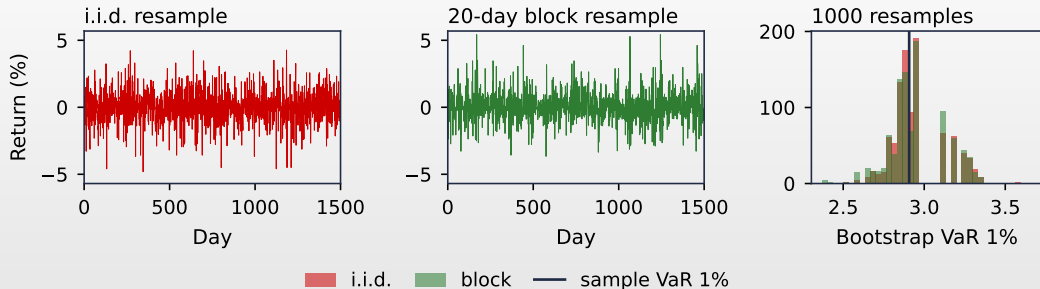
- Historical simulation (HS): no model, only the past losses sorted, $L_{(1)} \leq \dots \leq L_{(n)}$

$$\widehat{\text{VaR}}_{\alpha} = L_{(j)} + (p - j)(L_{(j+1)} - L_{(j)}),$$

$$p = 1 + (n - 1)(1 - \alpha), \quad j = \lfloor p \rfloor$$

- ▶ p : position among the sorted losses (np.quantile, linear interpolation); $j = \lfloor p \rfloor$: its integer part
- $\widehat{\text{ES}}_{\alpha}$: the mean of the k losses $\geq \widehat{\text{VaR}}_{\alpha}$, $k \approx n\alpha$
 - ▶ Chart: $n = 40$, $\alpha = 10\%$, $p = 36.1$, $k = 4$
 - ▶ The tail count $n\alpha$ drives precision: $n = 500$, $\alpha = 1\%$ leaves five losses
- HS is unconditional: a calm window gives a low VaR even if today is volatile

What You Need for Today (9/21): Bootstrap Intervals



- **Bootstrap Efron (1979):** draw n losses with replacement, recompute the statistic $\hat{\theta}^{(b)}$, repeat for $b = 1, \dots, B$
 - ▶ 95% percentile interval: $[q_{2.5\%}, q_{97.5\%}]$ of the $\hat{\theta}^{(b)}$; its width measures the sampling uncertainty
- i.i.d. draws break volatility clustering (left); the **moving-block bootstrap Künsch (1989)** draws blocks of, e.g., 20 consecutive days, concatenated and cut to length n (middle)
 - ▶ Right: with clustered crises, the block distribution of VaR 1% is wider than the i.i.d. one

What You Need for Today (10/21): Precision of VaR

- ▣ **Asymptotic standard error of HS VaR** (i.i.d. losses)

$$\text{SE}(\widehat{\text{VaR}}_\alpha) \approx \frac{\sqrt{\alpha(1-\alpha)/n}}{f(\text{VaR}_\alpha)}$$

- ▶ f : density of L at VaR; few observations near VaR (small f) = imprecise VaR; $\sqrt{\alpha(1-\alpha)/n}$: standard error of the share of losses above VaR
- ▶ $\hat{f}$: Gaussian kernel density, bandwidth (Silverman) $h = 0.9 \min(\hat{\sigma}, \text{IQR}/1.34) n^{-1/5}$, IQR: interquartile range

- ▣ **HAC** (heteroskedasticity- and autocorrelation-consistent) version: replace $\alpha(1-\alpha)$ by the long-run variance of $l_t = 1\{L_t > \widehat{\text{VaR}}\}$

$$\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j=1}^m \left(1 - \frac{j}{m+1}\right) \hat{\gamma}_j$$

- ▶ $\hat{\gamma}_j$: sample autocovariance of l_t at lag j ; Bartlett weights $1 - j/(m+1)$; Newey–West lag $m = \lfloor 4(n/100)^{2/9} \rfloor$; clustered exceedances give $\hat{\Omega} > \alpha(1-\alpha)$

What You Need for Today (11/21): Precision of ES

- **ES**: influence function [Chen \(2008\)](#) and its variance

$$\text{IF}(x) = \text{VaR}_\alpha + \frac{(x - \text{VaR}_\alpha)_+}{\alpha} - \text{ES}_\alpha, \quad \text{Var}(\widehat{\text{ES}}_\alpha) \approx \frac{\text{Var}(\text{IF}(L))}{n}$$

- ▶ $\text{IF}(x)$: the effect of one extra observation x on the estimate; only losses beyond VaR move it, each with weight $1/\alpha$
- ▶ So the precision of $\widehat{\text{ES}}_\alpha$ is driven by the few largest losses: the tail count $n\alpha$ and how heavy the tail is beyond VaR
- Rule of thumb: relative standard error of an i.i.d. ES estimate roughly $\propto 1/\sqrt{n\alpha}$
 - ▶ $n = 250$, $\alpha = 2.5\%$: only 6.25 tail losses; a block bootstrap widens the interval further when the tail days cluster

What You Need for Today (12/21): Filtered Historical Simulation

- FHS Barone-Adesi et al. (1999): fit an AR(1)-GARCH(1,1) (Chapter 5) by quasi-maximum likelihood (QML) and standardise the returns

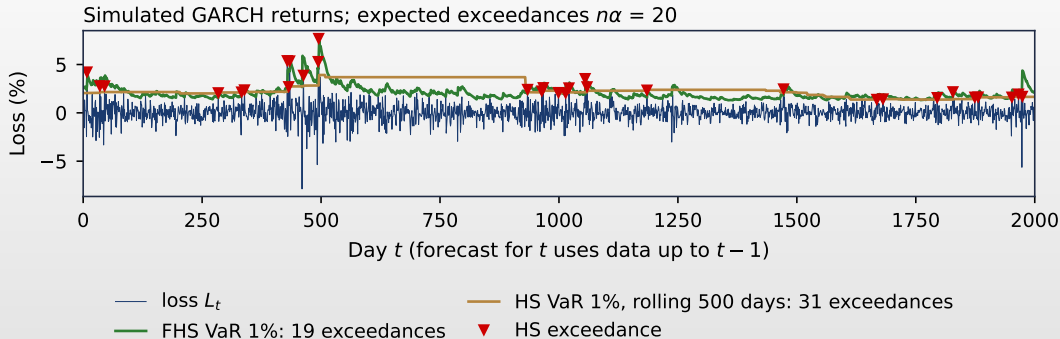
$$r_t = \mu_t + \sigma_t z_t, \quad \mu_t = c + \phi r_{t-1}, \quad \sigma_t^2 = \omega + \alpha_1 (r_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2, \quad \hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t}$$

- ▶ μ_t : conditional mean (AR(1): c constant, ϕ autoregressive coefficient); σ_t : conditional volatility; z_t : i.i.d. shock with mean 0, variance 1; $\hat{z}_t$: standardised residual
- ▶ $\omega > 0$; α_1 : reaction to yesterday's shock; β_1 : memory; $\alpha_1 + \beta_1 < 1$
- Tomorrow's VaR and ES: forecast scale times a residual tail statistic, minus the forecast mean

$$\text{VaR}_{t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{\xi}_\alpha, \quad \hat{\xi}_\alpha = -q_\alpha(\hat{z}), \quad \text{ES}_{t+1} = -\hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \overline{(-\hat{z})}_{\geq \hat{\xi}_\alpha}$$

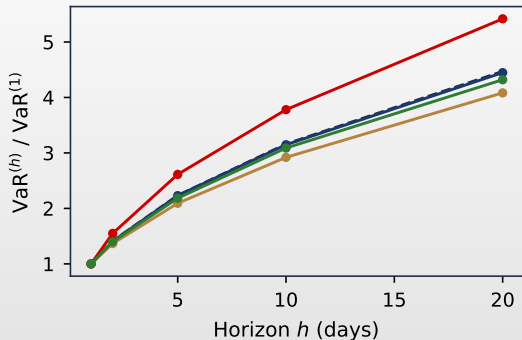
- ▶ $\overline{(-\hat{z})}_{\geq \hat{\xi}_\alpha}$: mean of the residual losses $-\hat{z}_t$ at or beyond $\hat{\xi}_\alpha$
- ▶ Example: $\hat{\mu}_{t+1} = 0.1\%$, $\hat{\sigma}_{t+1} = 3\%$, $\hat{\xi}_{1\%} = 2.8$: $\text{VaR}_{1\%} = -0.1 + 3 \times 2.8 = 8.3\%$

What You Need for Today (13/21): Filtered VaR and Exceedances



- Exceedance on day t : $L_t > \text{VaR}_t$, with VaR_t computed from data up to day $t - 1$ only (out of sample)
 - Under a correct model the count N over n days is $\text{Bin}(n, \alpha)$: mean $n\alpha$, standard deviation $\sqrt{n\alpha(1 - \alpha)}$; $n = 2000$, $\alpha = 1\%$: 20 ± 4.4
 - Chart (simulated GARCH): FHS follows the volatility and gives 19 exceedances; the rolling HS VaR reacts late and gives 31, grouped after volatility jumps

What You Need for Today (14/21): Longer Horizons



- $\sqrt{h}$ rule
- i.i.d. Normal
- i.i.d. Student-t(4)
- AR(1), $\phi = 0.2$
- GARCH

- Horizon of h days: log losses add up

$$L_t^{(h)} = L_t + L_{t-1} + \dots + L_{t-h+1}$$

- ▶ $\text{VaR}^{(h)}$: quantile of these sums; consecutive sums share $h - 1$ days (overlap)

- ▶ Simple returns compound:
 $100[1 - \prod_{i=1}^h (1 + R_{t+i}/100)]$, R in %

- Square-root-of-time rule: $\text{VaR}^{(h)} = \sqrt{h} \text{VaR}^{(1)}$

- ▶ Exact only for i.i.d. zero-mean Normal losses: the variance of a sum of h independent days is h times the daily one
- ▶ Chart (simulated): positive autocorrelation pushes the ratio above $\sqrt{h}$; fat tails pull it below

What You Need for Today (15/21): Peaks Over Threshold: the GPD

- Excesses $Y = L - u$ over a high threshold u follow approximately a **generalised Pareto distribution (GPD)** [McNeil, Frey & Embrechts \(2015\)](#)

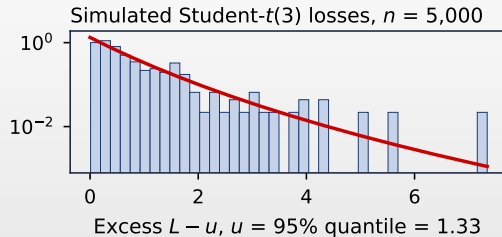
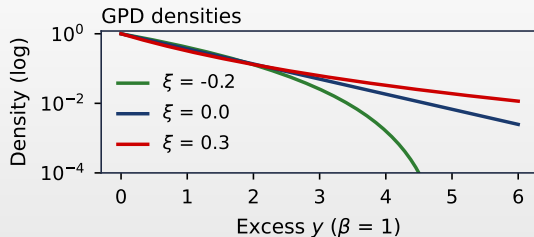
$$P(Y \leq y \mid L > u) \approx G_{\xi, \beta}(y) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y \geq 0$$

- ξ : shape; $\xi > 0$: heavy, power-law tail (moments of order $\geq 1/\xi$ are infinite); $\xi = 0$: exponential tail ($1 - e^{-y/\beta}$), as for the Normal distribution; $\xi < 0$: bounded tail; $\beta > 0$: scale
- Tail VaR and ES (n : sample size, N_u : number of exceedances, $\xi < 1$)

$$\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} \alpha \right)^{-\xi} - 1 \right], \quad \text{ES}_\alpha = \frac{\text{VaR}_\alpha + \beta - \xi u}{1 - \xi}$$

- N_u/n estimates $P(L > u)$; the GPD extrapolates beyond the largest observed losses, where HS has no data
 - i.i.d. standard errors of the maximum-likelihood estimates: $\text{SE}(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$, $\text{SE}(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$; choice of u : bias (too low) against variance (too high)

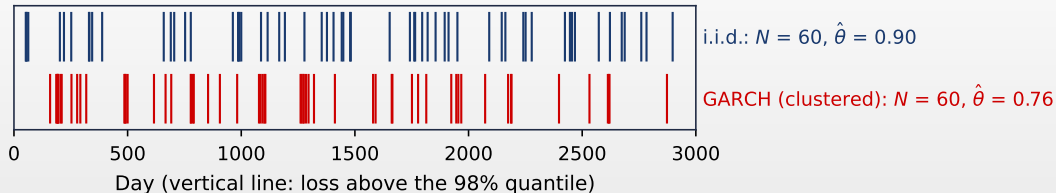
What You Need for Today (16/21): The GPD on Simulated Losses



250 excesses over u fitted GPD, $\hat{\xi} = 0.13$, $\hat{\beta} = 0.75$

- Left: GPD densities with $\beta = 1$ on a log scale: $\xi = -0.2$ ends at $y = 5$, $\xi = 0$ is a straight line (exponential), $\xi = 0.3$ decays as a power
- Right: 5 000 Student- $t(3)$ losses, threshold at the 95% quantile, 250 excesses; the fitted $\hat{\xi} = 0.13$ is below the true tail value $1/3$: with 250 excesses $SE(\hat{\xi}) \approx 0.07$, and a threshold that is not high enough biases $\hat{\xi}$

What You Need for Today (17/21): Clustered Exceedances: the Extremal Index

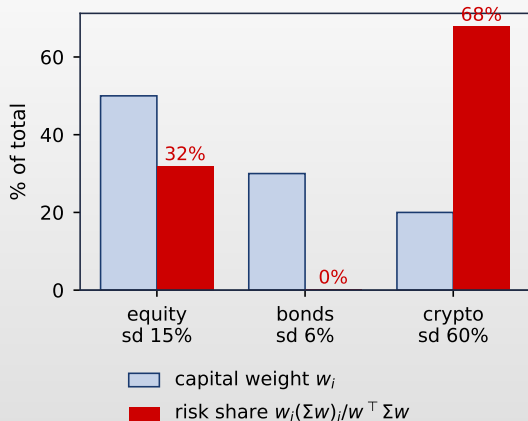


- **Extremal index** $\theta \in (0, 1]$: the reciprocal of the mean cluster size of exceedances; $\theta = 1$: isolated exceedances (i.i.d.), $\theta = 0.5$: clusters of two on average

$$\hat{\theta} = \min \left\{ 1, \frac{2 \left(\sum_i (T_i - 1) \right)^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} \right\} \quad \text{if } \max_i T_i > 2$$

- Intervals estimator [Ferro & Segers \(2003\)](#): $S_1 < \dots < S_N$: days of the N exceedances; $T_i = S_{i+1} - S_i$: gaps between them; the chart: exceedances of the 98% quantile, i.i.d. against GARCH
- About θN independent clusters: i.i.d. standard errors computed from N exceedances are too small when $\theta < 1$

What You Need for Today (18/21): Risk Contributions (Euler Allocation)



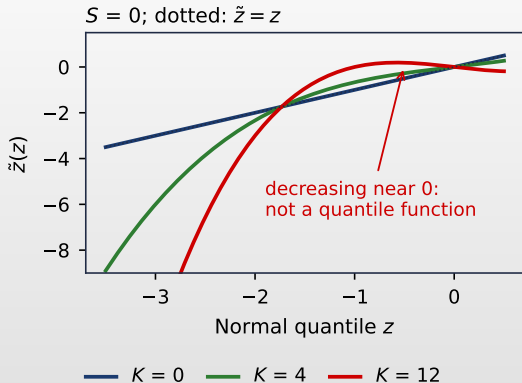
- Normal portfolio: exposures w , covariance matrix Σ , mean neglected

$$\sigma_p = \sqrt{w^T \Sigma w}, \quad \text{VaR} = z_{1-\alpha} \sigma_p,$$

$$\frac{\partial \text{VaR}}{\partial w_i} = z_{1-\alpha} \frac{(\Sigma w)_i}{\sigma_p}, \quad C_i = w_i \frac{\partial \text{VaR}}{\partial w_i}$$

- Marginal VaR $\partial \text{VaR} / \partial w_i$: change of VaR per extra unit in asset i ; component VaR C_i [Tasche \(2008\)](#):
 $\sum_i C_i = \text{VaR}$ (Euler: VaR is homogeneous of degree 1)
- ES contributions: $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}]$, the mean of asset i 's loss on the tail days of the portfolio
- Diversification benefit: $\sum_i z_{1-\alpha} w_i \sigma_i - \text{VaR} \geq 0$
 - Risk parity: weights with equal shares, $w_i(\Sigma w)_i = w_j(\Sigma w)_j$ for all i, j
 - Chart (hypothetical assets): a 20% weight in a volatile asset carries about two thirds of the risk

What You Need for Today (19/21): The Cornish–Fisher Expansion

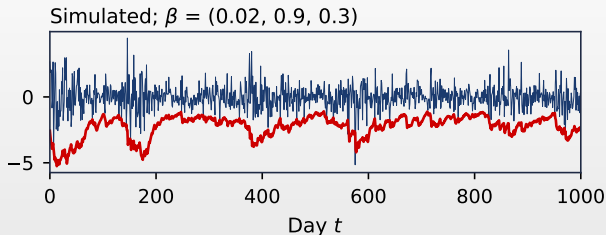
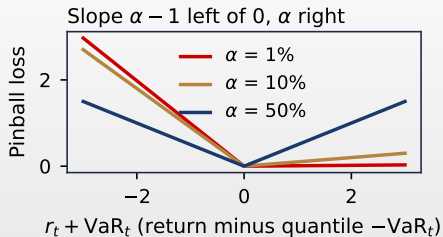


- Adjusts the Normal quantile for skewness S and excess kurtosis K [Cornish & Fisher \(1938\)](#)

$$\tilde{z} = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$$

- ▶ $S = \mathbb{E}[(X - \mu)^3]/\sigma^3$; $K = \mathbb{E}[(X - \mu)^4]/\sigma^4 - 3$ (0 for the Normal distribution); $\text{VaR}_\alpha = -(\mu + \sigma \tilde{z}(z_\alpha))$
 - ▶ $K > 0$ pushes the far-tail quantile further out; $S < 0$ lengthens the left tail
- A valid quantile function must be increasing in z ; chart ($S = 0$): with large K , $\tilde{z}$ decreases near 0, so the expansion stops being a quantile function

What You Need for Today (20/21): Quantile Regression, the Pinball Loss and CAViaR



— return r_t — $-\text{VaR}_t$, SAV: $\beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$

- **Pinball loss:** the loss whose expected value is minimised by the true α -quantile $-\text{VaR}_t$

$$\ell_\alpha(r_t, \text{VaR}_t) = (\alpha - 1\{r_t < -\text{VaR}_t\}) (r_t + \text{VaR}_t)$$

- ▶ An exceedance costs $(1 - \alpha)$ per unit, a day above the quantile costs α per unit (left); a lower average pinball loss = better VaR forecasts
- **CAViaR SAV** [Engle & Manganelli \(2004\)](#): VaR follows its own recursion, fitted by minimising $\sum_t \ell_\alpha$
 - ▶ $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$; $|\beta_1| < 1$: persistence; β_2 : reaction to yesterday's absolute return (right, simulated)

What You Need for Today (21/21): Comparing Forecasts and Regulatory Capital

- **Diebold–Mariano test** [Diebold & Mariano \(1995\)](#): are two forecasts equally accurate?

$$d_t = \ell_t^A - \ell_t^B, \quad DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}_d/n}} \approx N(0, 1) \text{ under } H_0 : \mathbb{E}[d_t] = 0$$

- ▶ ℓ_t^A, ℓ_t^B : losses (e.g. pinball) of models A and B on day t ; $\bar{d}$: mean difference; $\hat{\Omega}_d$: HAC long-run variance of d_t ;
 $DM < -1.96$: A significantly better at 5%

- **Joint VaR–ES loss FZ0** [Patton, Ziegel & Chen \(2019\)](#): scores a pair $(v_t, e_t) = (\text{return quantile, tail mean})$, $e_t \leq v_t < 0$

$$\text{FZ0} = -\frac{1\{r_t \leq v_t\} (v_t - r_t)}{\alpha e_t} + \frac{v_t}{e_t} + \ln(-e_t) - 1$$

- **FRTB-style capital** (C2, simplified) [Basel Committee \(2019\)](#): ES from a stressed 250-day window, scaled to the liquidity horizon

$$\text{Capital} = m_c \times \text{ES}_{10}^{\text{stres}} \times \sqrt{LH/10} \times \text{amount}$$

- ▶ $m_c = 1.5$: multiplier; $\text{ES}_{10}^{\text{stres}}$: 10-day ES 2.5% in the worst window; LH : liquidity horizon in days (10 or 20)

Notation and Conventions Used in Every Exercise

- Return X (in %), loss $L = -X$; level $\alpha =$ tail probability; VaR and ES are **positive** numbers when they are losses (definitions: [What You Need \(1/21\)](#))
- Empirical convention (all B exercises): `np.quantile(L, 1 - alpha)` and the tail mean of [What You Need \(8/21\)](#)
 - ▶ $\widehat{ES}_\alpha$ averages $k \approx n\alpha$ losses: an approximation of the exact ES, equal to it when $n\alpha$ is an integer
- Units: log losses in % of the position for one asset; simple returns for portfolios (they add across assets); percentage points (pp) for standard errors and interval widths
- Symbols that look alike: $\alpha =$ tail probability, $\alpha_1 =$ GARCH coefficient; $\xi =$ GPD shape, $\hat{\xi}_\alpha =$ residual quantile ([B3](#)); $w =$ money exposures ([A5](#)) or portfolio weights ([B5](#))

A1 [Solved]: Normal VaR and ES — Task

▣ **Question:** what are VaR and ES of a position whose daily return is Normal?

▣ **Data**

- ▶ $X \sim N(\mu, \sigma^2)$, daily log return in % of the position, $\mu = 0.04\%$, $\sigma = 1.2\%$; loss $L = -X$ with density f
- ▶ $z_\alpha = \Phi^{-1}(\alpha) < 0$; Φ , φ : distribution and density functions of the standard Normal distribution

▣ **Tasks**

1. Derive $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$ from $ES_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$.
2. Compute z_α and $\varphi(z_\alpha)$ at $\alpha = 1\%$ and $\alpha = 2.5\%$.
3. Compute VaR and ES at both levels, in % of the position.

▣ **Report:** a four-cell table (VaR, ES $\times$ 1%, 2.5%)

▣ **Interpretation**

- ▶ which of the four numbers are almost equal?
- ▶ why does that matter for regulation?

▣ Notebook: section [A1](#)

A1 [Solved]: Solution

- **1. Derivation.** Write $X = \mu + \sigma Z$; $X \leq q_\alpha$ is $Z \leq z_\alpha$, of probability α

► Since $\varphi'(u) = -u\varphi(u)$:

$$\int_{-\infty}^z u\varphi(u) du = [-\varphi(u)]_{-\infty}^z = -\varphi(z)$$

► $\mathbb{E}[X \mid X \leq q_\alpha] = \mu + \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = \mu - \sigma \frac{\varphi(z_\alpha)}{\alpha};$

change the sign

- **2–3.** $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$, e.g.
 $-(0.04 + 1.2 \cdot (-2.326)) = 2.75$;
 $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16$

α	z_α	$\varphi(z_\alpha)$	φ/α	VaR_α	ES_α
1%	-2.326	0.0267	2.665	2.75	3.16
2.5%	-1.960	0.0584	2.338	2.31	2.77

- VaR and ES in % of the position; for a position of 1 million currency units: VaR 1% $\approx 27\,516$, ES 1% $\approx 31\,583$
- Interpretation: ES 2.5% (2.77%) $\approx$ VaR 1% (2.75%): under the Normal distribution the switch to ES 2.5% keeps the capital, which is why FRTB chose this level

A2 [Solved]: Student- t ES — Task

▣ **Question:** how much do heavy tails change VaR and ES when the standard deviation is the same as in [A1](#)?

▣ **Data**

► $X = m + sT$, $T \sim t_\nu$ with density g_ν , $t_\alpha = t_\nu^{-1}(\alpha) < 0$; $\nu = 4$, $m = 0.04\%$, standard deviation 1.2%

► Tail integral (proof in Extension Pack 1): $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu + t^2}{\nu - 1}$ for $\nu > 1$

▣ **Tasks**

1. Use the tail integral to write $ES_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$.
2. Find the scale s that gives a standard deviation of 1.2%, using $\text{Var}(T) = \nu/(\nu - 2)$ (finite only for $\nu > 2$).
3. Compute VaR and ES at 1% and 2.5% and put them next to [A1](#).

▣ **Report:** the [A1](#) table with a Student- t column

▣ **Interpretation**

- at which level do the two models differ most?
- why does ES see the difference first?

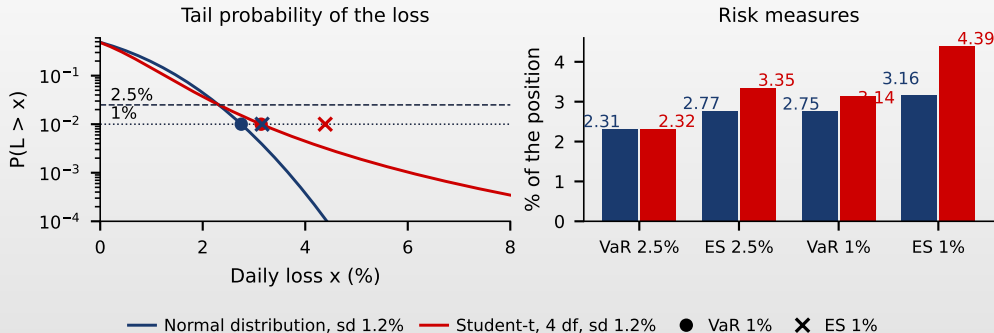
▣ Common mistake: using $\sigma = 1.2$ instead of s as the scale

A2 [Solved]: Solution

- 1. Divide the tail integral by α and rescale with $X = m + sT$, as in A1
- 2. $s^2 \cdot 4/2 = 1.2^2$, so $s = 1.2\sqrt{2/4} = 0.849$
- 3. At 1%: $t_{1\%} = -3.747$, $g_4(t_{1\%}) = 0.0087$, ES factor $\frac{0.0087}{0.01} \cdot \frac{4 + 14.04}{3} = 5.221$
 - $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14$; $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39$

	$\text{VaR}_{2.5\%}$	$\text{ES}_{2.5\%}$	$\text{VaR}_{1\%}$	$\text{ES}_{1\%}$
Normal	2.31	2.77	2.75	3.16
t_4	2.32	3.35	3.14	4.39

A2 [Solved]: Solution — Chart



- Interpretation: VaR 2.5% is the same in both models; VaR 1% and both ES are much higher under t_4 : heavy tails show only deep in the tail, and ES sees them first

A3 [Solved]: VaR and ES of a Discrete Loss — Task

- **Question:** what are VaR and ES when the loss has an atom (a default)?
- **Data:** a bond loses 60 monetary units (loss given default, LGD) with probability $p = 3\%$ and 0 otherwise
- **Tasks**
 1. Compute VaR 5% and VaR 2.5% from the infimum definition of VaR ([What You Need \(1/21\)](#)).
 2. Compute ES 5% and ES 2.5% with the formula for a loss with an atom ([What You Need \(7/21\)](#)) [Acerbi & Tasche \(2002\)](#).
 3. Explain why $ES_{5\%}$ differs from $\mathbb{E}[L \mid L \geq VaR_{5\%}]$.
- **Report:** four numbers and one sentence

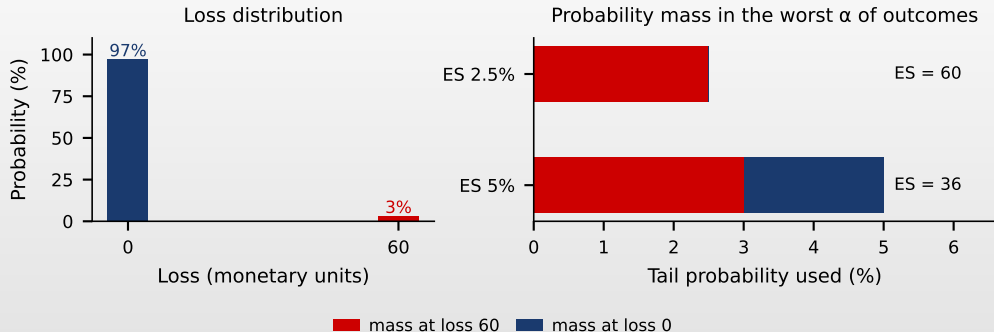
Loss L	0	60
$P(L)$	97%	3%
$P(L \geq \cdot)$	100%	3%

- The worst α of outcomes must be filled with probability mass; at $\alpha = 5\%$ the 3% default mass is not enough

A3 [Solved]: Solution

- 1. $P(X \leq -60) = 3\% \leq 5\%$, $P(X \leq 0) = 100\% > 5\%$: $q_{5\%} = 0$, $\text{VaR}_{5\%} = 0$; $P(X \leq -60) = 3\% > 2.5\%$: $\text{VaR}_{2.5\%} = 60$
- 2. $\text{ES}_{5\%} = \frac{60 \cdot 0.03 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$; $\text{ES}_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$
- 3. $\mathbb{E}[L \mid L \geq 0] = 1.8$ averages over **all** outcomes; ES 5% uses the 3% default mass plus 2% of the zero-loss atom (chart)
- Interpretation: this fractional mass is the exact version of the tail average used in B1 with $k \approx n\alpha$ losses

A3 [Solved]: Solution — Chart



A4 [Proposed]: When VaR Punishes Diversification — Task

- **Question:** can a VaR limit make a diversified portfolio look riskier than its parts?
- **Data:** two independent bonds A and B; each loses 100 with probability $p = 0.7\%$ (the lecture used 0.9%) and 0 otherwise; portfolio A + B [Artzner et al. \(1999\)](#)
- **Tasks**
 1. Write the distribution of the portfolio loss: the values 0, 100, 200 and their probabilities.
 2. Compute VaR 1% and ES 1% of one bond and of the portfolio with the discrete formula of [A3](#).
 3. Show that VaR 1% is not subadditive here while ES 1% is.
 4. Repeat steps 2–3 at $\alpha = 1.5\%$.
- **Report:** a table of VaR and ES for A, B and A + B at both levels
- **Interpretation:** what would a VaR-based limit reward?
- **Worked example to follow:** [A3](#)

B1 [Solved]: Historical VaR and ES on BET — Task

▣ Questions

- ▶ how large are the tail losses of the Romanian stock market?
- ▶ how precise are the estimates?

▣ Data

- ▶ BET index, daily closes 2000-01-05 – 2026-09-18, weekdays only; $n = 6\,670$ daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L_t = -r_t$
- ▶ Two samples: the full period and the last 500 returns (from 2024-09-12, shaded in the chart)

▣ Tasks

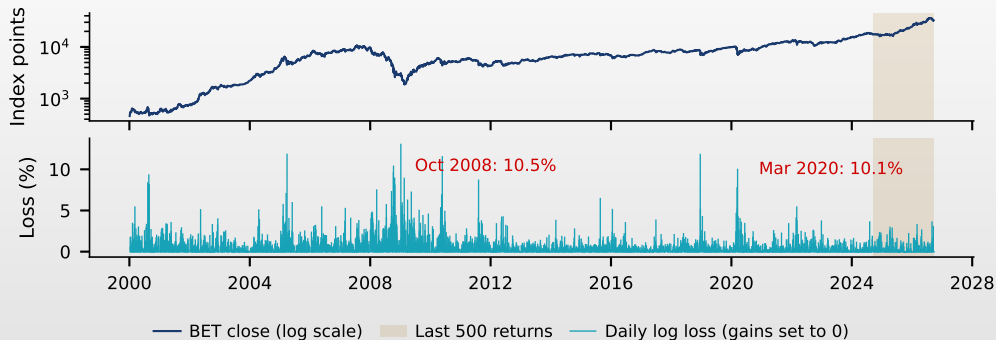
1. Compute historical VaR and ES at 1% and 2.5% on both samples, and count the losses beyond VaR 1%.
2. Build a 95% percentile bootstrap interval for VaR 1% with 2 000 i.i.d. resamples, seed 42 [Efron \(1979\)](#).
3. Repeat step 2 with a moving-block bootstrap: 20-day blocks drawn with replacement, concatenated and cut to length n [Künsch \(1989\)](#).

▣ Report: a table of the eight VaR/ES numbers and the two intervals per sample

▣ Interpretation

- ▶ which estimate would you give a risk committee?
- ▶ why is the block interval wider on the full sample?

B1 [Solved]: Historical VaR and ES on BET — Data



□ Look first: the largest losses (October 2008: 10.45%; March 2020: 10.08%) come in clusters

B1 [Solved]: Solution (1/4) — Step by Step

- **1. Last 500 days.** Sort the losses; VaR 1% sits at position $1 + 499 \cdot 0.99 = 495.01$; $L_{(i)}$: the i -th smallest loss

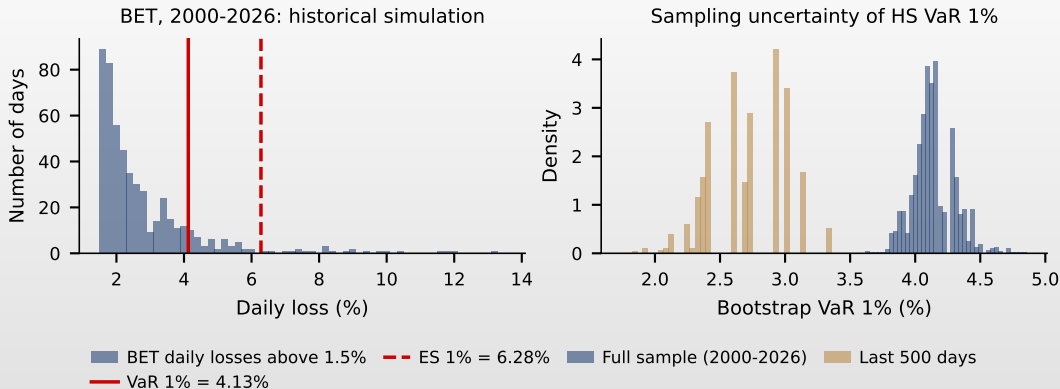
► $\widehat{\text{VaR}}_{1\%} = L_{(495)} + 0.01 (L_{(496)} - L_{(495)}) = 2.710 + 0.01 (2.925 - 2.710) = 2.712\%$

► The five losses beyond it: 3.68, 3.35, 3.12, 3.03, 2.93: their mean is $\widehat{\text{ES}}_{1\%} = 3.22\%$

	n	$\text{VaR}_{1\%}$	$\text{ES}_{1\%}$	$\text{VaR}_{2.5\%}$	$\text{ES}_{2.5\%}$	beyond VaR 1%
2000–2026	6 670	4.13	6.28	2.80	4.57	67
Last 500 days	500	2.71	3.22	2.05	2.69	5

- The 500-day ES 1% averages only five numbers: report it with that count
- Both are unconditional summaries: the full sample describes 26 years, the last 500 days a calm period; tomorrow's risk needs a conditional forecast (B3)

B1 [Solved]: Solution (2/4) — The Tail and the Bootstrap

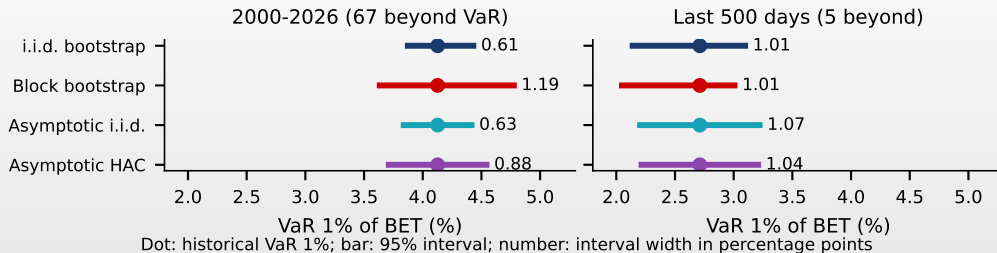


B1 [Solved]: Solution (3/4) — The Tail and the Bootstrap

- Left: BET losses above 1.5% with historical VaR 1% and ES 1%; right: the 2000 i.i.d. bootstrap values of VaR 1% for each sample
- Each resample recomputes both the quantile and which losses lie beyond it; the 2.5% and 97.5% percentiles of the 2000 values give the interval
- The two bootstrap distributions do not overlap: the recent market is calmer than the 26-year average

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B1 [Solved]: Solution (4/4) — Intervals



- 2. i.i.d. bootstrap: full [3.85, 4.46], last 500 days [2.11, 3.12]
- 3. Block bootstrap: [3.61, 4.80] and [2.02, 3.03]

- Interpretation: exceedances cluster in 2008 and 2020, so blocks that keep a crisis together widen the full-sample interval by about half (in B1 Extended: asymptotic and HAC); the calm last 500 days show no such effect

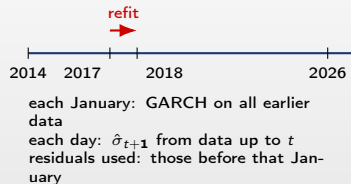
B2 [Proposed]: How Precise Is an ES Estimate? — Task

- **Question:** does a 500-day historical ES 2.5% pin the tail risk down to within ± 0.3 pp [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)?
- **Data**
 - ▶ S&P 500 (daily close) and BET, daily log losses in %, 2000–2026; samples: full and last 500 days
 - ▶ Use the resampling function of [B1](#) (2 000 resamples, seed 42, 20-day blocks) with ES 2.5% as the statistic
- **Tasks**
 1. Compute ES 2.5%, the tail count $n\alpha$ and the number k of losses that the tail mean actually averages.
 2. Draw one i.i.d. and one block resample of the S&P 500 last 500 days, and recompute VaR 2.5%, the tail membership and ES on each.
 3. Build the i.i.d. and the block 95% intervals for all four samples and compare their widths with 0.6 pp (± 0.3).
- **Report:** a table ($n\alpha$, k , ES, two intervals) and an interval chart
- **Interpretation**
 - ▶ which samples meet the ± 0.3 pp target?
 - ▶ what governs the width?
- **Worked example to follow:** [B1](#)

B3 [Proposed]: Tomorrow's Filtered VaR for Bitcoin — Task

- **Question:** Bitcoin's VaR 1% for 19 September 2026: filtered or historical?
- **Data**
 - ▶ Bitcoin (USD), daily close, 17 September 2014 – 18 September 2026, 7 days a week; daily log returns in %
 - ▶ AR(1)-GARCH(1,1) by quasi-maximum likelihood (QML): $r_t = \mu_t + \sigma_t z_t$ [Barone-Adesi et al. \(1999\)](#)
- **Tasks**
 1. Estimate the model on the full sample and forecast $\hat{\mu}_{t+1}$, $\hat{\sigma}_{t+1}$.
 2. Compute $\hat{\xi}_{1\%} = -q_{1\%}(\hat{z})$ from the standardised residuals and the filtered VaR 1% and ES 1%.
 3. Compare with historical VaR and ES on the last 500 days and on the full sample.
 4. With the forecasts of the timeline (right), count the days of the last two years with a loss above the FHS VaR and above a rolling 500-day HS VaR.
- **Report:** three VaR/ES pairs, two exceedance counts
- **Worked example to follow:** [B1](#)

Forecast timeline (step 4)



- No forecast uses information from its own day or later: a full-sample threshold is not an out-of-sample forecast
- **Interpretation**
 - ▶ why do the three VaRs differ?
 - ▶ which method misses more often?

C1 [Proposed]: Critique an AI Answer — the Answer

- **Prompt:** "px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive log losses in %, and the 10-day VaR 1%."
- **Answer:** the code below, which runs without an error message

```
1  import numpy as np
2  r = 100 * np.log(px).diff().dropna()           # daily log returns, %
3  loss = -r                                       # log loss, %
4  var1 = np.percentile(r, 99)                   # VaR 1%
5  es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
6  var10 = 10 * var1                             # 10-day VaR 1%
```

- **Conclusion of the AI:** "VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46%."

C1 [Proposed]: Critique an AI Answer — Task

□ Questions

- ▶ which lines of the AI answer are wrong?
- ▶ what are the correct numbers?

□ **Data:** the BET data of B1; the answer contains **three** mistakes typical for VaR and ES, everything else is correct

□ Tasks

1. For each mistake, name the line number and explain in one sentence why it is wrong, using $\text{VaR}_\alpha = -q_\alpha(r) = q_{1-\alpha}(L)$.
2. Say how you would detect it: a count of observations, a number from B1, or a slide of the lecture.
3. Write the corrected lines, including the 10-day VaR from overlapping 10-day losses, and recompute the three numbers.
4. Check the corrected ES 2.5%: how many losses does it average? How many should it average?

□ **Report:** an error table (line, error, detection, correction) and the corrected numbers; log each error you actually found in `AI_ERRORS.md`

□ **Interpretation:** why does the code run without an error message although the answer is wrong?

□ **Worked example to follow:** B1

B4 [Solved]: A GPD Tail for BET — Task

▣ **Question:** what does an extreme-value tail add beyond historical simulation, deep in the tail?

▣ **Data**

- ▶ The BET losses of B1 ($n = 6\,670$)
- ▶ Peaks over threshold McNeil, Frey & Embrechts (2015): the GPD with shape ξ and scale β , its standard errors and the tail VaR and ES (formulas: What You Need (15/21)); N_u = number of exceedances

▣ **Tasks**

1. Set u = the empirical 95% quantile of L and form the excesses $L - u$ of the losses above it.
2. Fit the GPD by maximum likelihood with location fixed at 0: `scipy.stats.genpareto.fit(excesses, floc=0)`.
3. Compute the i.i.d. standard errors $SE(\hat{\xi})$ and $SE(\hat{\beta})$.
4. Compute GPD VaR and ES at 1%, 0.5% and 0.1% and compare with historical simulation.
5. Refit for thresholds leaving 10% down to 1.5% of the losses above them and plot $\hat{\xi}$ with 95% bands.

▣ **Report:** the fit with standard errors, a six-row VaR/ES table with tail counts, and the stability chart

▣ **Interpretation**

- ▶ is the BET tail heavier than the Normal distribution?
- ▶ how sure are we?

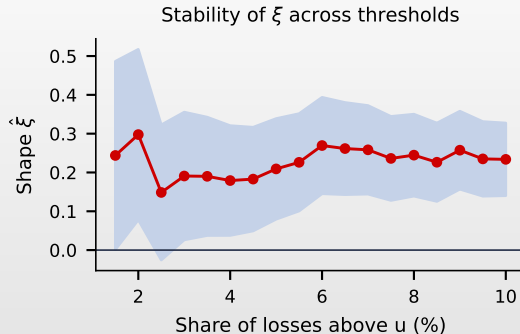
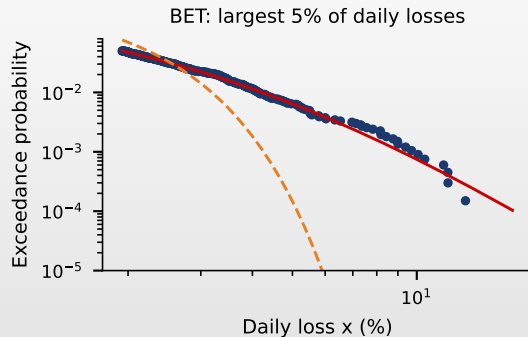
B4 [Solved]: Solution (1/3) — Fit and Table

- 1–2. $u = 1.93\%$ leaves $N_u = 334$ of $n = 6\,670$ losses above it; $\hat{\xi} = 0.209$, $\hat{\beta} = 1.20$
- 3. $SE(\hat{\xi}) = (1 + 0.209)/\sqrt{334} = 0.066$, $SE(\hat{\beta}) = 0.10$; $\hat{\xi}/SE = 3.2$
- 4. VaR 1%: $N_u/n = 0.050$, $(0.01/0.050)^{-0.209} = 1.401$
 - ▶ $VaR_{1\%} = 1.93 + \frac{1.20}{0.209}(1.401 - 1) = 4.22\%$

	VaR GPD	VaR hist.	ES GPD	ES hist.
1%	4.22	4.13	6.34	6.28
0.5%	5.47	5.38	7.92	7.86
0.1%	9.18	9.50	12.61	11.25

- Losses beyond historical VaR: 67 at 1%, 34 at 0.5%, 7 at 0.1%
- The two methods agree down to 0.5%; at 0.1% historical ES averages 7 numbers, while the GPD uses all 334 excesses through a fitted tail

B4 [Solved]: Solution (2/3) — Diagnostics



● Empirical $P(L \geq x)$ — GPD fit, $\xi = 0.21$ — Normal distribution 95% confidence band — Estimated ξ

B4 [Solved]: Solution (3/3) — Diagnostics

- Left: the GPD follows the empirical tail, the Normal tail falls far below it; right: $\hat{\xi}$ ranges from 0.15 to 0.30 across thresholds, while VaR 0.1% moves only between 9.04% and 9.51%
- Interpretation: the tail is heavier than the Normal one, but the i.i.d. SE assumes independent exceedances; BET losses cluster, so the honest uncertainty is larger (B4 Extended: extremal index)

A5 [Solved]: Risk Contributions of Two Assets — Task

▣ **Question:** does each asset carry risk in proportion to the money invested in it?

▣ **Data**

- ▶ Money exposures $w = (600\,000, 400\,000)$ EUR in equity and bonds; daily simple returns R in decimals, standard deviations $\sigma_1 = 0.012$, $\sigma_2 = 0.009$, correlation $\rho = -0.2$
- ▶ $\Sigma = \begin{pmatrix} 0.000144 & -0.0000216 \\ -0.0000216 & 0.000081 \end{pmatrix}$; (Σ : covariance matrix) jointly Normal returns, mean ignored, $\alpha = 1\%$, $z_{99\%} = 2.326$ (the 99% standard Normal quantile)

▣ **Tasks**

1. Compute the portfolio standard deviation σ_p in EUR and the portfolio VaR 1% (formulas: [What You Need \(18/21\)](#)).
2. Compute Σw , the marginal VaR per EUR and the component VaR C_i ; check $C_1 + C_2 = \text{VaR}$ [Tasche \(2008\)](#).
3. Compute the diversification benefit: the sum of stand-alone VaRs minus the portfolio VaR.
4. Under the Normal distribution, rescale each C_i by $\frac{\varphi(z_{1\%})/0.01}{z_{99\%}}$ to get the ES 1% contributions.

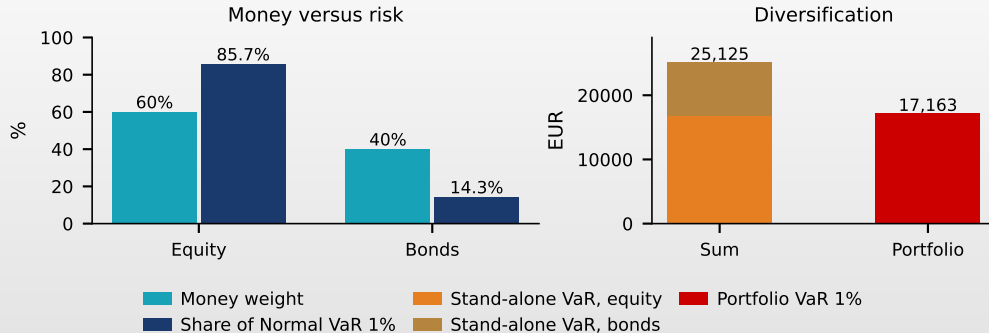
▣ **Report:** a table of C_i , C_i^{ES} and shares, and the diversification benefit in EUR

▣ **Interpretation:** do the bonds carry risk in proportion to their 40% money weight?

A5 [Solved]: Solution

- 1. $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$, $\sigma_p = 7\,378$ EUR, $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$ EUR
- 2. $\Sigma w = (77.76, 19.44)$; marginal VaR per EUR (0.0245, 0.0061); $C_1 = 14\,711$ (85.7%), $C_2 = 2\,452$ (14.3%), sum 17 163 EUR
- 3. Stand-alone $16\,750 + 8\,375 = 25\,125$ EUR vs 17 163 EUR
- 4. Factor 1.146: $C_1^{ES} = 16\,854$, $C_2^{ES} = 2\,809$, total 19 663 EUR = ES 1%; same shares as VaR under the Normal distribution

A5 [Solved]: Solution — Chart



□ Interpretation: the bonds hold 40% of the money but 14.3% of the risk: the negative correlation makes them a partial hedge

B5 [Proposed]: Risk Contributions of a Four-Asset Portfolio — Task

- **Question:** in an equally weighted portfolio of SPY, TLT, GLD and Bitcoin, who carries the risk?
- **Data**
 - ▶ SPY, TLT, GLD (adjusted closes) and Bitcoin (USD closes), 18 September 2014 – 18 September 2026; prices aligned on the 3017 common days, then daily simple returns R in %
 - ▶ Fractional weights $w_i = 25\%$, rebalanced daily; Σ = sample covariance of R (in $\%^2$); Normal VaR 1% with the mean ignored; historical ES 2.5%
- **Tasks**
 1. Compute the Normal VaR 1%, the marginal VaR and the component VaR of each asset, as in [A5](#).
 2. Compute the historical ES 2.5% of the portfolio loss $L_p = -w^\top R$ and the contributions $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$ as averages over the tail days.
 3. Compute the undiversified VaR $\sum_i z_{99\%} w_i \sigma_i$.
- **Report:** a table of weights, contributions and shares, and a chart of money weights vs risk shares
- **Interpretation:** is equal money equal risk? (comparing Normal VaR 1% with historical ES 2.5% changes both the model and the level)
- **Worked example to follow:** [A5](#)

Exit Check and Team Project

Exit check: can you say why

- a 500-day ES 1% should be reported with the number 5?
- $\sqrt{10}$ scaling is too low for BET?
- a 25% position can carry 80% of the risk (B5, after the lecture)?
- portfolio losses need simple, not log returns?

Team project

- Choose one of C2 or C3 (Extension Pack 4), with the common rubric
- Log AI use in AI_USE.md and every error you actually caught in AI_ERRORS.md (no quota)

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Tail algebra	A2 Extended [Solved], A4 Extended, A6, A7, A5 Extended [Solved], A8	A1–A5
2. Inference	B1 Extended [Solved], B3 Extended, B6, B5 Extended	B1, B3, B5
3. Extremes and dynamics	B4 Extended [Solved], B7	B3, B4
4. Projects	C2 (ES capital, BVB vs S&P 500), C3 (deep quantile regression); rubric	the lecture case study, B1–B3

- Each exercise has a notebook section with the same label and names its worked example; do them in order within a pack

A2 Extended [Solved]: The Student- t Tail Integral — Task and Solution

- **Question:** where does the Student- t ES factor of A2 come from?
- **Data:** $g_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2}$, the t_ν density; c_ν : the constant that makes it integrate to 1
- **Tasks**
 1. Differentiate $(1 + x^2/\nu)^{-(\nu+1)/2}$.
 2. Deduce $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu+t^2}{\nu-1}$ and state where $\nu > 1$ is needed.
- **Report:** the two-line proof

Solution

- 1. $\frac{d}{dx}(1 + x^2/\nu)^{-(\nu+1)/2} = -\frac{\nu+1}{\nu} x (1 + x^2/\nu)^{-(\nu+1)/2}$, so $x g_\nu(x) = -\frac{\nu c_\nu}{\nu-1} \frac{d}{dx}(1 + x^2/\nu)^{-(\nu+1)/2}$
- 2. Integrating from $-\infty$ to t : $-\frac{\nu}{\nu-1} g_\nu(t)(1 + t^2/\nu) = -g_\nu(t) \frac{\nu+t^2}{\nu-1}$; the term at $-\infty$ vanishes only if $\nu > 1$
- ES needs $\nu > 1$; matching a variance needs $\nu > 2$

A4 Extended [Proposed]: For Which Levels Does VaR Fail? — Task

- ▣ **Question:** for which tail probabilities α is VaR not subadditive in A4?
- ▣ **Data:** the A4 bonds, $p = 0.7\%$, and the VaR definition of What You Need (1/21)
- ▣ **Tasks**
 1. Write $\text{VaR}_\alpha(A)$ and $\text{VaR}_\alpha(A + B)$ as step functions of α .
 2. Find the interval of α where $\text{VaR}_\alpha(A + B) > 2\text{VaR}_\alpha(A)$ and explain both endpoints.
- ▣ **Report:** the interval and one sentence per endpoint
- ▣ **Worked example to follow:** A4

A6 [Proposed]: When Is Cornish–Fisher a Valid Quantile? — Task

▣ **Question:** for which excess kurtosis is the Cornish–Fisher map still a quantile function?

▣ **Data**

- ▶ $\mu = 0$, $\sigma = 1.2\%$, skewness $S = -0.5$, excess kurtosis K [Cornish & Fisher \(1938\)](#)
- ▶ The Cornish–Fisher map $\tilde{z}(z)$ of [What You Need \(19/21\)](#); $\text{VaR}_\alpha = -\sigma\tilde{z}(z_\alpha)$

▣ **Tasks**

1. With $K = 3$, compute the three correction terms at $z_{1\%}$, $\tilde{z}_{1\%}$ and the Cornish–Fisher VaR 1%.
2. Write $d\tilde{z}/dz = Az^2 + Bz + C$ and locate the vertex of this parabola.
3. Find the largest K for which $\tilde{z}$ is increasing on $[z_{0.1\%}, 0]$, and decide whether $K = 10$ is admissible.

▣ **Report:** the VaR, the bound on K and a plot of $\tilde{z}$ and $d\tilde{z}/dz$ for $K = 3$ and $K = 10$

▣ **Interpretation:** does a valid quantile map also give an accurate VaR?

▣ **Worked example to follow:** [A1](#)

A7 [Proposed]: ES as a Minimum, and Subadditivity — Task

- ▣ **Question:** why can ES never reward splitting a portfolio?
- ▣ **Data**
 - ▶ Integrable loss L ; the objective $G_\alpha(v)$ [Rockafellar & Uryasev \(2000\)](#) of [What You Need \(6/21\)](#)
 - ▶ Numbers: the [A4](#) portfolio (two bonds, $p = 0.7\%$) at $\alpha = 1\%$
- ▣ **Tasks**
 1. For the [A4](#) portfolio, compute the slopes of G on $(0, 100)$ and $(100, 200)$, the minimiser and the minimum.
 2. Show that the right derivative of G is $1 - P(L > v)/\alpha$ and deduce that VaR_α minimises G , with minimum ES_α .
 3. Deduce $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$ from $(a + b)_+ \leq a_+ + b_+$.
- ▣ **Report:** the three numbers of step 1 and two short proofs
- ▣ **Worked example to follow:** [A3](#), [A4](#)

A5 Extended [Solved]: ES Contributions from the Minimum — Task and Solution

- **Question:** why are the historical ES contributions of B5 averages over the tail days?
- **Data:** $ES_\alpha(w) = \min_v G(w, v)$,
 $G(w, v) = v + \mathbb{E}[(L_p - v)_+]/\alpha$, $L_p = -w^\top R$ (A7)
- **Tasks**
 1. Use the envelope argument at $v^* = \text{VaR}_\alpha$ to compute $\partial ES_\alpha / \partial w_i$.
 2. Show that $w_i \partial ES_\alpha / \partial w_i = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha]$ for a continuous L_p .
- **Report:** a short proof
- **Worked example to follow:** A7

Solution

- $\partial G / \partial v = 0$ at v^* , so
 $\partial ES_\alpha / \partial w_i = \partial G / \partial w_i(w, v^*) = \mathbb{E}[-R_i \mathbf{1}\{L_p > v^*\}]/\alpha$
- For continuous L_p , $P(L_p > v^*) = \alpha$: this is $\mathbb{E}[-R_i \mid L_p \geq \text{VaR}_\alpha]$; summing $w_i \times$ it gives ES_α (Euler)
- Historical estimate: average $-w_i R_i$ over the tail days (B5); under the Normal distribution it equals the VaR rescaling of A5

A8 [Proposed]: The Sampling Variance of Historical ES — Task

- ▣ **Question:** how imprecise is a 500-day historical ES 2.5%, even for Normal losses?
- ▣ **Data**
 - ▶ n i.i.d. losses, continuous, positive density at VaR_α , $\mathbb{E}[L^2] < \infty$; $Y = (L - \text{VaR}_\alpha)_+$, the excess of the loss over VaR (0 below it)
 - ▶ The influence function IF [Chen \(2008\)](#) and $\text{Var}(\widehat{\text{ES}}_\alpha) \approx \text{Var}(\text{IF}(L))/n$ of [What You Need \(11/21\)](#)
- ▣ **Tasks**
 1. Show that $\text{Var}(\text{IF}(L)) = \text{Var}(Y)/\alpha^2$ and explain, with [A7](#), why estimating VaR adds no first-order term.
 2. For $L \sim N(0, 1)$, $\alpha = 2.5\%$, integrate, with $v = \text{VaR}_\alpha = -z_\alpha$, $\mathbb{E}[Y] = \int_v^\infty (x - v)\varphi(x) dx$ and $\mathbb{E}[Y^2]$ in closed form.
 3. Compute the SE of $\widehat{\text{ES}}_{2.5\%}$ for $n = 500$ and $\sigma = 1.2\%$, and check it by simulation.
- ▣ **Report:** the two moments, the SE in pp and the simulated standard deviation
- ▣ **Interpretation:** why is the S&P 500 value ([B2](#)) much larger?
- ▣ **Worked example to follow:** [A1](#), [A7](#)

B1 Extended [Solved]: Asymptotic and HAC Intervals for VaR — Task

- **Question:** do the asymptotic formulas agree with the bootstrap of B1?
- **Data**
 - ▶ The BET losses of B1; the i.i.d. and HAC standard errors of What You Need (10/21) (kernel density $\hat{f}$, Newey–West lag $\lfloor 4(n/100)^{2/9} \rfloor$)
- **Tasks**
 1. Normal benchmark (from A1): with $f(\text{VaR}_{1\%}) = \varphi(z_{1\%})/1.2$, compute the SE for $n = 500$ and $n = 6\,670$.
 2. For BET, compute $\hat{f}$, the i.i.d. and HAC SEs and the two 95% intervals on both samples.
- **Report:** the four intervals of VaR 1% next to the two bootstrap intervals of B1
- **Interpretation:** why are the BET intervals wider than the Normal benchmark?

B1 Extended [Solved]: Solution

- ▣ 1. $f = 0.0267/1.2 = 0.0222$: SE 0.20 pp ($n = 500$, width 0.79), 0.055 pp ($n = 6\,670$, width 0.22)
- ▣ 2. BET full: $\hat{f} = 0.0076$, SE 0.16 (i.i.d.), 0.23 (HAC); intervals [3.81, 4.44], [3.69, 4.57]
- ▣ Last 500 days: $\hat{f} = 0.0163$, SE 0.27 and 0.27
- ▣ Interpretation: the BET tail is heavier than the Normal one, so the density at VaR is lower and the quantile less precise; HAC and blocks agree that clustering widens the full-sample interval
- ▣ Chart of the four intervals per sample: [B1](#), Solution — Intervals

B3 Extended [Proposed]: A Bootstrap Interval for Tomorrow's FHS VaR — Task

- ▣ **Question:** how uncertain is the filtered VaR of B3?
- ▣ **Data:** the B3 model and data; residual bootstrap Christoffersen & Gonçalves (2005) with 999 paths, seed 42, function `fhs_bootstrap` of the seminar Quantlet
- ▣ **Tasks**
 1. For each path, simulate an AR(1)-GARCH(1,1) series of the sample length with resampled $\hat{z}$ and re-estimate $\theta^* = (c^*, \phi^*, \omega^*, \alpha_1^*, \beta_1^*)$.
 2. Compute μ_{t+1}^* and σ_{t+1}^* by running the **original** Bitcoin returns through θ^* , and $\hat{\xi}_{1\%}^*$ from the path's own residuals.
 3. Report the 90% percentile interval of $\text{VaR}^* = -\mu_{t+1}^* + \sigma_{t+1}^* \hat{\xi}_{1\%}^*$ and the intervals of its two sources.
- ▣ **Report:** the histogram of the 999 forecasts with the interval
- ▣ **Interpretation**
 - ▶ does the interval include the 500-day historical VaR?
 - ▶ is that a test?
- ▣ **Worked example to follow:** B3

B6 [Proposed]: Cornish–Fisher vs Empirical Quantiles — Task

- ▣ **Question:** does the Cornish–Fisher correction fix the Normal VaR on real data?
- ▣ **Data**
 - ▶ BET (compulsory): the [B1](#) data; extension: S&P 500, Bitcoin and gold (daily closes, weekdays for gold) and the BNR EUR/RON reference rate (2005–2026), all to 18 September 2026
 - ▶ Moments of the returns X : mean, standard deviation with `ddof=1`, `scipy.stats.skew` and `scipy.stats.kurtosis` (excess, biased)
- ▣ **Tasks**
 1. Compute S and K for BET.
 2. Compute VaR 1% and VaR 2.5% by the Normal distribution, Cornish–Fisher ([A6](#)) and historical simulation.
 3. Build 95% i.i.d. bootstrap intervals (1 000 resamples) for the two historical VaRs and check whether the Cornish–Fisher values fall inside.
 4. Separately, check whether K exceeds the monotonicity bound of [A6](#).
- ▣ **Report:** a small table per asset and a chart of the three methods with the bootstrap intervals
- ▣ **Interpretation:** why does Cornish–Fisher work better at 2.5% than at 1%?
- ▣ **Worked example to follow:** [A1](#), [A6](#), [B1](#)

B5 Extended [Proposed]: Risk-Parity Weights — Task

- ▣ **Question:** which long-only, fully invested weights give each asset of B5 the same share of Normal VaR?
- ▣ **Data:** the B5 returns and covariance Σ
- ▣ **Tasks**
 1. Solve $w_i(\Sigma w)_i = w_j(\Sigma w)_j$ for all i, j , with $w \geq 0$ and $\sum_i w_i = 1$, numerically.
 2. Compare with the inverse-volatility weights $w_i \propto 1/\sigma_i$.
 3. At the risk-parity weights, compute the Normal VaR 1%, the historical ES 2.5% and the ES shares.
- ▣ **Report:** the weights, VaR, ES and shares
- ▣ **Interpretation:** does equal VaR risk mean equal ES risk?
- ▣ **Worked example to follow:** B5

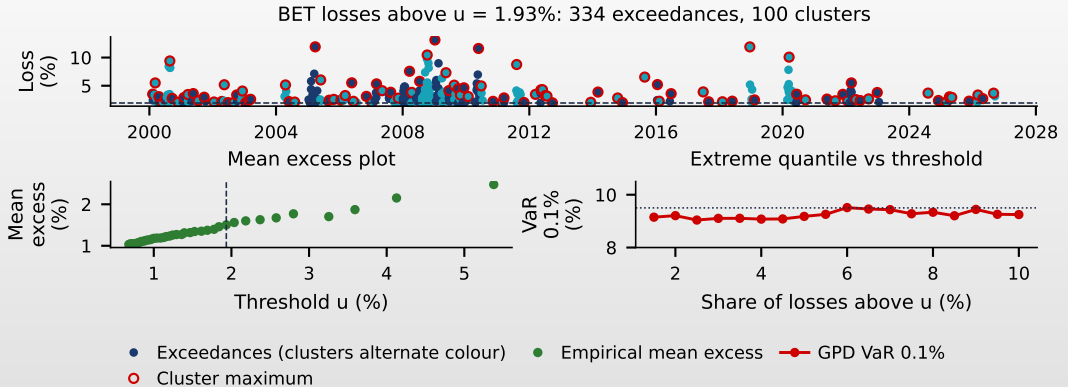
B4 Extended [Solved]: Clustered Exceedances and the Extremal Index — Task

- ▣ **Question:** how many independent pieces of tail information do the 334 BET exceedances of B4 contain?
- ▣ **Data**
 - ▶ Exceedance times $S_1 < \dots < S_N$ of u (B4), gaps $T_i = S_{i+1} - S_i$
 - ▶ The intervals estimator $\hat{\theta}$ Ferro & Segers (2003) of What You Need (17/21)
- ▣ **Tasks**
 1. Compute $\hat{\theta}$ and the target number of clusters $C = \lfloor \hat{\theta}N \rfloor + 1$.
 2. Split the exceedances at the $C - 1$ largest gaps; if the $(C - 1)$ -th and C -th largest gaps are equal, reduce C until they differ.
 3. Fit the GPD to the cluster maxima and compare $\hat{\xi}/\text{SE}$ with B4.
- ▣ **Report:** $\hat{\theta}$, the number of clusters, the declustered fit and a timeline chart
- ▣ **Interpretation:** is the i.i.d. SE of B4 credible?

B4 Extended [Solved]: Solution

- ▣ 1. $\hat{\theta} = 0.32$, $C = 108$
- ▣ 2. Tie at the cut ($T = 12$ days): 100 clusters of 3.3 exceedances on average; largest cluster 33, 47 isolated exceedances
- ▣ 3. $\hat{\xi} = 0.269$ (SE 0.127) vs 0.209 (SE 0.066); $\hat{\xi}/\text{SE}$ falls from 3.2 to 2.1; at the VaR 1% threshold $\hat{\theta} = 0.40$
- ▣ Interpretation: the tail stays heavy, but the i.i.d. SE is not credible; report the declustered fit or filter with GARCH first (conditional EVT)

B4 Extended [Solved]: Solution — Chart



B7 [Proposed]: CAViaR on BET — Task

- **Question:** does modelling the BET quantile directly beat filtered and historical simulation out of sample?
- **Data**
 - ▶ The BET returns of [B1](#) ($n = 6\,670$)
 - ▶ SAV CAViaR [Engle & Manganelli \(2004\)](#) and the pinball loss of [What You Need \(20/21\)](#); $\text{VaR}_1 = -\hat{q}_1\%$ of the first 300 returns; routines `caviar_path`, `caviar_fit` (seminar Quantlet): 10^4 starts in $(0, 1)^3$, seed 42, the best 10 refined
- **Tasks**
 1. Estimate β on 2000–2015 by minimising the summed pinball loss at $\alpha = 1\%$.
 2. Compute one pinball-loss term step by step for the first out-of-sample day and check that the forecast for day t uses only r_{t-1} .
 3. With β fixed, forecast VaR for 2016–2026 and compare exceedances and mean pinball loss with FHS (refitted each January) and rolling 500-day HS.
 4. For 20 February – 30 April 2020 count exceedances and report the peak VaR of each method.
- **Report:** the estimates, a comparison table and a chart with a COVID-19 zoom
- **Interpretation:** can the difference be attributed to the model class when CAViaR is fitted once and FHS every year?
- **Worked example to follow:** [B3](#); lecture, CAViaR on the S&P 500

C2 [Open]: ES Capital, BVB vs S&P 500 — Task (1/2)

- ▣ **Question:** which portfolio needs more ES capital per unit invested: Bucharest blue chips or the S&P 500?
- ▣ **Data**
 - ▶ BVB basket: Banca Transilvania, OMV Petrom, BRD–Groupe Société Générale, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica (adjusted closes), equal weights, rebalanced daily; days on which no stock traded are removed
 - ▶ S&P 500 portfolio: SPY, the S&P 500 ETF (adjusted closes); both from 2016-09-20 to 18 September 2026; daily simple returns R_t in % in local currency
- ▣ **Tasks**
 1. Construction: compute the one-day historical ES 2.5% of each portfolio (loss $-R_t$) and its tail count.
 2. Horizon: compute the 10-day ES 2.5% by $\sqrt{10}$ scaling, from overlapping compounded losses $100[1 - \prod_{i=1}^{10}(1 + R_{t+i}/100)]$, and by filtered Monte Carlo (AR(1)-GARCH(1,1) on log returns, today's state, 100 000 paths, seed 42, each path converted to a simple loss).
- ▣ **Report:** continued on the next slide
- ▣ **Worked example to follow:** [B1](#), [B3](#); lecture, horizon and Basel sections

C2 [Open]: ES Capital, BVB vs S&P 500 — Task (2/2)

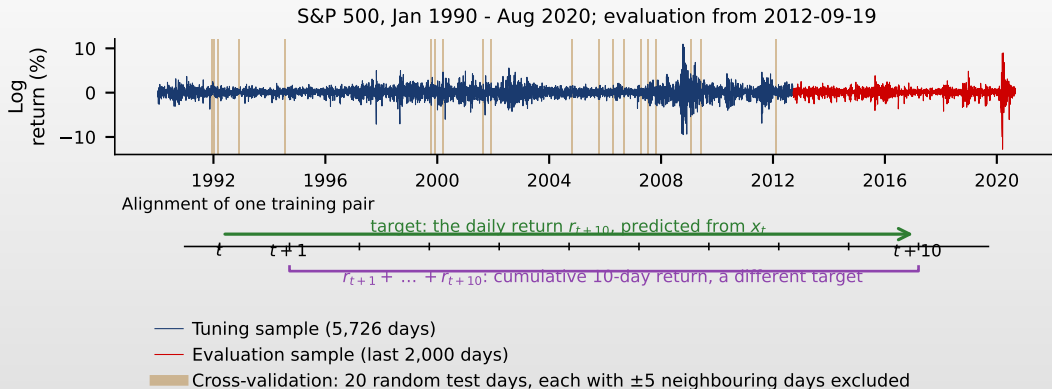
▣ Tasks (continued)

3. Stress: find the 250 consecutive days with the largest one-day historical ES 2.5%, plot the rolling ES with the window shaded, and report the stressed ES at 1 day and, by $\sqrt{10}$, at 10 days.
 4. Capital: capital per 1 000 000 invested = $1\,000\,000 \times 1.5 \times \text{ES}_{10}^{\text{stress}} (\%) / 100$ (1.5: the FRTB multiplier m_c ; $\text{ES}_{10}^{\text{stress}}$: the 10-day stressed ES), for a 10-day and a 20-day liquidity horizon (a further factor $\sqrt{2}$); amounts in RON for BVB and in USD for SPY, ratios comparable.
 5. Uncertainty: 95% moving-block bootstrap intervals (20-day blocks, 2 000 resamples) for the one-day ES and for the stressed ES with the window held fixed.
- ▣ **Report:** a table of all numbers with intervals and a one-page argument
- ▶ which portfolio needs more capital?
 - ▶ how do thin trading and currency change the answer?
- ▣ A simplified illustration inspired by FRTB [Basel Committee \(2019\)](#), not the full capital calculation of [Basel Committee, MAR33](#)

C3 [Open]: Deep Quantile Regression for VaR — Task (1/2), Minimum Version

- **Question:** does a small neural network beat linear quantile regression for the S&P 500 VaR 1%, as in the lecture case study [Chronopoulos, Raftapostolos & Kapetanios \(2024\)](#)?
- **Data**
 - ▶ S&P 500 daily closes, 1990-01-03 – 31 August 2020 (the paper starts in 1985); unscaled daily log returns r_t ; tuning: first 5 726 days, evaluation: last 2 000 (from 2012-09-19)
 - ▶ One predictor set: GARCH(1,1) σ_t (estimated on the tuning sample only); $\alpha = 1\%$
- **Tasks**
 1. Build training pairs (x_t, r_{t+10}) whose targets all lie in the tuning sample, as in the chart; the target is the daily return r_{t+10} , not the cumulative 10-day return.
 2. Fit a linear quantile regression and one network (one hidden ReLU layer of 5 units, ridge penalty 0.01, Adam optimiser (stochastic gradient descent with adaptive step sizes), early stopping) by the pinball loss; standardise the predictors on the training part of each cross-validation split only.
 3. Compare mean pinball losses on the evaluation sample with a Diebold–Mariano test (HAC, 10 lags) [Diebold & Mariano \(1995\)](#), over 5 random seeds.
- **Report:** a table of mean pinball losses, DM statistics and exceedance counts, per seed

C3 [Open]: Deep Quantile Regression for VaR — Data



C3 [Open]: Deep Quantile Regression for VaR — Task (2/2), Extensions

- ▣ **Replication extension:** the four predictor sets and the cross-validation grid of the paper (Appendix of the lecture); relative RMSFE only to reproduce Table 6, since RMSFE is minimised by the conditional mean, not the quantile
- ▣ **Rolling extension:** re-estimate once a year on a rolling window and forecast September 2020 – 2026 for the S&P 500 and the BET
- ▣ **Joint extension:** two outputs (v_t, e_t) trained on the FZ0 loss [Patton, Ziegel & Chen \(2019\)](#), (v_t : return quantile, e_t : tail mean, as in the lecture) with $e_t = -\exp(a_t)$ and $v_t = e_t + \exp(b_t)$, a_t, b_t unconstrained network outputs, so that $e_t < 0$ and $e_t \leq v_t$
- ▣ **Interpretation** (one page): can a gain be attributed to non-linearity when architecture, penalty and tuning change together?

Project Choice and Common Rubric

Choose one

- C2: ES capital for a portfolio of your choice (at least three assets not covered in class), with the four-step structure of C2
- C3: the minimum version, then one extension
- Deliver a Quantlet folder (code, `Metainfo.txt`, charts) and three slides: numbers, interpretation, limitations

Rubric

- Reproducibility: data loaded by code, fixed seeds, every number from a script (30%)
- Correct conventions: sign, level, horizon, return type (20%)
- Uncertainty: tail counts and intervals for every tail estimate (25%)
- Interpretation and limits, including logged AI use and the AI errors actually found (25%)

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